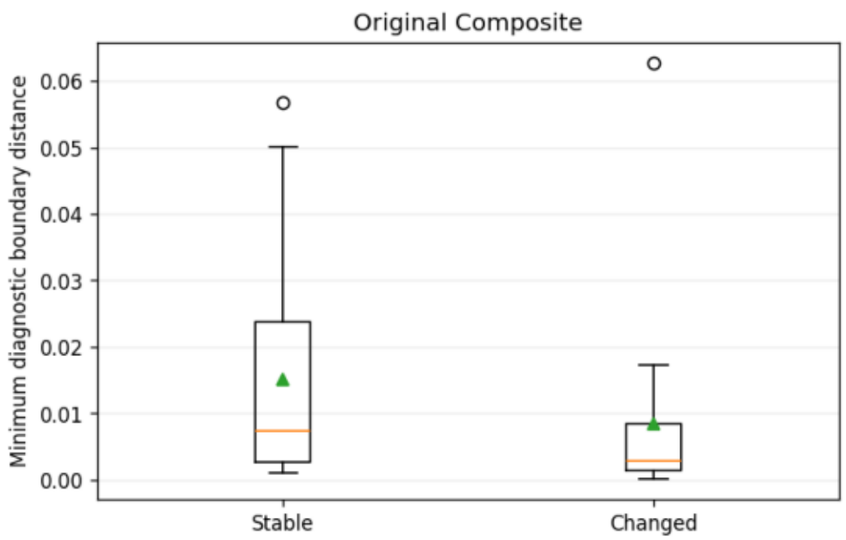
Ate Makoce — Agricultural Sustainability Assessment

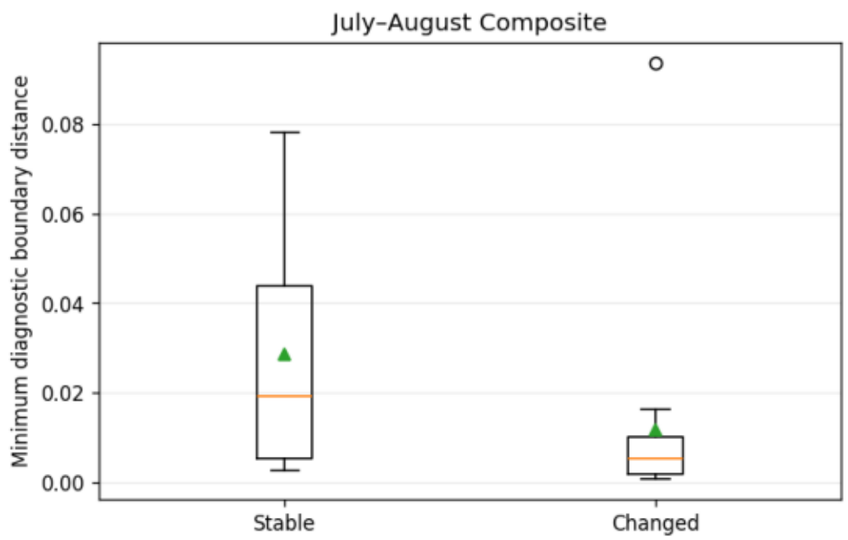
Environmental Data Cube for Oglala Lakota | Candidate Stability and Landsat Vegetation Indicators

(A) Candidate Stability — Original Composite



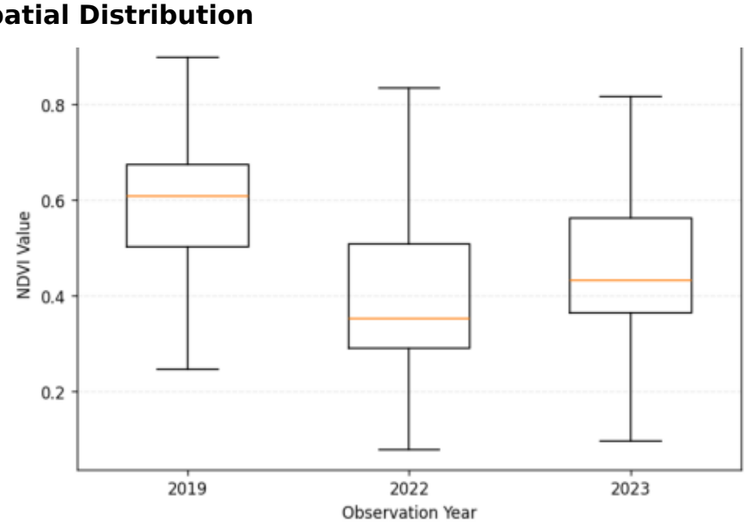
Stable candidates generally have greater diagnostic boundary distances and a wider distribution than Changed candidates. Overlap remains between groups.

(B) Candidate Stability — July–August Composite



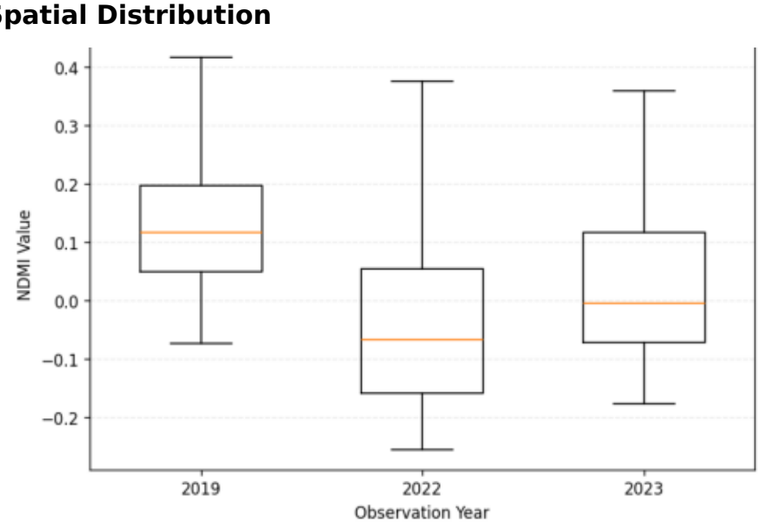
The summer composite shows greater variability among Stable candidates. Differences in boundary distance describe classification sensitivity, not accuracy.

(C) NDVI Spatial Distribution



Median vegetation greenness is highest in 2019, lower in 2022, and partially recovers in 2023. The distributions describe parcel-scale variability.

(D) NDMI Spatial Distribution



Median moisture-related spectral values are positive in 2019, lower in 2022, and closer to zero in 2023. These patterns do not independently establish drought.

Agricultural sustainability interpretation: The candidate stability analyses characterize sensitivity to diagnostic classification boundaries, while NDVI and NDMI describe changes in vegetation greenness and moisture-related spectral conditions across observation years. Together, these results provide complementary evidence for evaluating rangeland conditions. They do not independently establish overgrazing, land degradation, or causal effects of land management. Interpretation requires consideration of climate variability, grazing information, and additional field or management evidence.

Source: Notebook 09, SA 16.17.5E and SA 18.2. Saved original assessment figures; no analyses rerun.